

SOFTWARE DEVELOPMENT AGREEMENT & REQUIREMENTS DOCUMENT

1. Agreement Overview

This Agreement is made between:

Client: Muhammad Faiq Khan

Developer: Hamza Sheikh

The purpose of this agreement is the development of a Recommendation System for Universities and Programs.

2. Project Title

AI-Based Dynamic Recommendation System for Universities & Programs

3. Project Objective

To design and develop an intelligent recommendation system that suggests universities and academic programs to users based on their profile, behavior, and engagement data.

The system will dynamically improve its recommendations using user interactions and feedback signals.

4. System Description

The system will operate in two phases:

4.1 Onboarding-Based Personalization

After user signup, the system collects the following preferences:

- Degree level (e.g., Bachelors, Masters, etc.)
- City preference
- Field of interest (e.g., Computer Science, Business, etc.)
- Affordable fee range
- Previous academic marks/percentage range
- Intended start time/session

And are saved in the users table under the field *onboarding_preferences* that will be used for the recommendations.

4.2 Behavioral-Based Learning

The system continuously improves recommendations using user interactions such as:

- Click-through behavior (CTR on universities/programs)
- Search queries
- Application submissions
- Time spent on listings (if tracked)

These signals will be used to refine ranking scores over time.

4.3 University Weighting System

Certain universities that have MOUs (Memorandums of Understanding) will receive a controlled ranking advantage in recommendations. Which in database is (scholarbee-verified)

This will be implemented as a weighted boost factor in the ranking algorithm, ensuring fair visibility without fully overriding relevance.

5. System Nature

- Fully dynamic system
- Self-improving through data accumulation

6. Deliverables

The developer will deliver:

1. Complete End-to-End Working System

- Recommendation engine
- Backend logic

2. Technical Documentation

- System architecture
- Algorithm explanation
- Data flow diagram
- API or module description

3. Demo Video (detailed including technical explanation as well)

- Explanation of system workflow
- Live or simulated demonstration
- Explanation of recommendation logic and ranking

7. Demo Consideration Clause

Since recommendation systems require user interaction data to function optimally:

- A cold-start simulation dataset will be used for demonstration
- Synthetic user profiles and interactions will be created
- Pre-filled data will simulate real-world usage

- System will demonstrate ranking, filtering, and MOU-based boosting clearly
This ensures the demo remains functional and understandable even without live users.

8. Timeline

- Development Deadline: 30th May
- Live Demo: 31st May

9. Ownership & Restrictions

Strict conditions apply:

- The developer cannot claim ownership of the system after submission
- The developer cannot publish, distribute, or reuse any part of the system
- The developer cannot include this project in portfolio, CV, GitHub, or any public/private showcase
- All intellectual property, codebase, documentation, and materials will remain the sole property of Muhammad Faiq Khan
- The developer agrees to maintain confidentiality regarding all project details

10. Acceptance Clause

By agreeing to this document, both parties confirm:

- They understand and accept the scope of work
- They agree to the timeline and payment terms
- They agree to the ownership and confidentiality conditions

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Client: Muhammad Faiq Khan

Signature: _____


Date: _____ 5/24/26 _____

Developer: Hamza Sheikh

Signature: _____


Date: _____ 5/24/26 _____